

SHREEDHAR JOSHI

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GitHub | LinkedIn | Codeforces Profile

EDUCATION

Indian Institute of Information Technology (IIIT), Surat

2025 – Expected 2029

B.Tech - Electronics and Communication Engineering

GPA (2nd Sem): 8.05/10.0

- **Core Courses:** Data Structures & Algorithms, C Programming, Discrete Mathematics.

Indian Institute of Technology (IIT), Madras

Jul 2025 – Present

BS - Data Science and Applications

- **Core Focus:** Computational Thinking, Mathematical Foundations, Statistics, Machine Learning, Business Data Mgt.

Khyati World School

2023 – 2025

Class XII (Higher Secondary, CBSE)

Result: 86.6%

Kendriya Vidyalaya No.1 Shahibaug, Ahmedabad

2021 – 2023

Class X (Secondary, CBSE)

Result: 96.2%

TECHNICAL SKILLS

- **Languages:** C (Primary), Python, TypeScript, JavaScript, Embedded C, HTML/CSS, SQL (PostgreSQL).
- **Frameworks & Libraries:** React.js, Tailwind CSS v4, @dnd-kit, Node.js, NumPy.
- **Developer Tools & Software:** Docker, ClamAV, DBMS, Android Studio, Proteus, Git/GitHub, Arduino IDE, Atmel AVR.
- **Platforms & Deployment:** Supabase (BaaS), Render, GitHub Pages, Netlify.
- **Hardware & Prototyping:** Arduino Microcontrollers, Sensor Interfacing, Basic Circuit Design.
- **Interests:** Competitive Programming, Astronomy, Robotics, Digital Audio Production (BandLab).

EXPERIENCE & LEADERSHIP

Atmiya University (e-Yantra, IIT Bombay Initiative) | Summer Intern – Robotics & Embedded Systems Jun 2026 – Jul 2026

- Programmed microcontroller firmware utilizing **Embedded C** and worked with bare-metal registers on **Atmel AVR** chips.
- Designed and successfully simulated complex circuits and peripheral interfaces using **Proteus**.
- Collaborated at the Center of Excellence to interface analog/digital sensors and process live hardware feedback loops.

ASTRA (Official Space Sciences Club, IIIT Surat) | Founder & Student Co-ordinator Apr 2026 – Present

- Established the first astrophysics community at IIIT Surat; authored the organization's charter and framework.
- Spearheaded recruitment and successfully executed the inaugural orientation for incoming engineering student members.

TECHNICAL PROJECTS

SafeGate (Browser-Assisted Cloud Sandbox) | Python, Docker, ClamAV, PostgreSQL

May 2026

- Engineered a secure virtualization utility designed to safely analyze and inspect suspicious binaries and file downloads.
- Built isolated containment environments using **Docker** to spin up dynamic, secure runtimes for analysis verification.
- Integrated **ClamAV** to automate local signature-based malware heuristics checks and generate transient report states.

Northstar Kanban Board (Real-Time Task Management) | React, TypeScript, Supabase, Tailwind v4 May 2026

- Engineered a collaborative agile workspace utilizing **Supabase Realtime Channels** to achieve (<50ms) state synchronization.
- Designed an asynchronous **6-second undo transaction buffer** for optimistic layout UI state mutations, decoupling user interactions from immediate database mutations to reduce accidental data deletion.
- Authored complex backend **PostgreSQL triggers and automated logging functions** to stream real-time activity metrics.
- Optimized large-scale rendering bottlenecks by applying aggressive memoization (`React.memo`) and callback caching.

AI Contract Risk Scanner | *Python, React, GitHub, TypeScript*

Feb 2026

- Engineered an automated parsing tool to translate complex legal contracts into simplified risk-weighted summaries.
- Implemented pattern-matching logic to flag high-risk legal clauses, improving contract readability.

HONORS & AWARDS

- **State Topper | Indian Olympiad Qualifier in Junior Science (NSEJS):** Achieved a Top 1% State Rank in the National Standard Examination; recognized for proficiency in Physics, Chemistry, and Biology (May 2022).
- **INSPIRE Awardee (DST) - MANAK:** Recipient of the Innovation in Science Pursuit for Inspired Research national award and a grant of INR 10,000 from the Department of Science and Technology, Govt. of India (Jan 2023).